Fresh aromatic herbs containing methylchavicol did not exhibit the pro-oxidative effects of pure methylchavicol on a human hepatoma cell line, HepG2.

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Methylchavicol (CH(3)-CV), an important aromatic constituent of different plants like tarragon and basils, has been shown to be carcinogenic by a mechanism yet unclear, although it has been reported that carcinogenicity of CH(3)-CV in rodent might be linked to its metabolic conversion into a genotoxic electrophilic metabolite generated through a two steps bioactivation pathway catalyzed by cytochrome P450 enzymes and sulfotransferases. The induction of carcinogenesis by certain agents has been associated with the generation of oxidative stress. The aim of the present study was to determine whether pure methylchavicol applied on a human hepatoma cell line, HepG2, could promote oxidative stress and might alter the expression of procarcinogenic biomarkers such as the drug-metabolizing enzyme (CYP2E1), the inducible form of nitric oxide synthase (iNOS) and might induce the expression of Cu/Zn-superoxide dismutase (Cu/Zn-SOD) and Mn-SOD that control the redox equilibrium of the cells. CH(3)-CV was shown to cause a significant induction of oxidative stress, as revealed by luminol-dependent chemiluminescence (LDCL) and to alter dramatically the expression of CYP2E1, iNOS and Mn-SOD, indicating that the toxic effect of CH(3)-CV could be mediated through a nitric oxide dependent mechanism. Under similar experimental conditions, the extracts from tarragon, chervil and basil did not induce such biological changes. These results provide evidence that the generation of an oxidative stress may be a significant event occurring during CH(3)-CV-induced toxicity. It also suggests that natural extracts containing different amounts of CH(3)-CV (tarragon, chervil and basil) did not elicit such toxicity and might contain compounds able to counteract this detrimental property.